

Mixed Model Diagnostics

Model↔Data Agreement

- Model attempts to describe how data were generated in order to learn about process
 - “All models are wrong, but some are useful” -G.E. Box
- Important to check model-data agreement
 - Do the data violate model assumptions?
 - Should model components be refined?
 - Remove or add predictors/covariate
 - Alter covariance structure
 - Are results sensitive to model choice or data?

Influence

- The ability of a single data point or multiple data points, through their presence or absence, to
 - alter important aspects of the analysis
 - yield qualitatively different inferences
 - violate model assumptions
- Want to determine these points to hopefully develop a better, more robust, model

Common Checks for Classical Linear Models

- Residual analysis
 - Residual plots
 - Histograms / QQplots
- Goodness of fit
 - Likelihood ratio test
 - Model selection criteria (e.g., AIC / BIC)
- Influence analysis
 - Cook's D
 - DFFITS

Linear Mixed Model

$$Y = X\beta + Z\gamma + \varepsilon$$

- Y is $n \times 1$ vector of observations
- X is an $n \times p$ fixed effects design matrix
- Z is an $n \times g$ random effects design matrix

$$\varepsilon \sim N(0, R) \text{ and } \gamma \sim N(0, G)$$

Model Agreement Issues

- Both $\hat{\beta}$ and $\hat{\gamma}$ depend on covariance parameter estimates
- Consider mixed model in both conditional and unconditional (marginal) sense
 - $Y \mid \gamma$ condition on particular values of random effects
 - Y unconditional, average over random effects
- Observation might influence fixed effects given covariance parameters, covariance parameters, or both

Assessment Strategy

- Similar to handling repeated measures data
 - Influence of single observation
 - Influence of set of observations from same EU
- **Issue:** Cannot determine effect of deleted observation(s) without refitting the model unless you assume covariance unaffected
- Traditional OLS summaries can give counter-intuitive results

Residuals

- Marginal Residual

$$r_i^m = y_i - x_i' \hat{\beta}$$

- Conditional Residual

$$r_i^c = y_i - x_i' \hat{\beta} - z_i' \hat{\gamma}$$

- MIXED produces these with the OUTPM= and OUTP= options respectively

Standardized Residuals

- Have options to also obtain studentized, Pearson, and scaled residuals.

Type	Marginal	Conditional
Raw	$r_i^m = y_i - x_i' \hat{\beta}$	$r_i^c = y_i - x_i' \hat{\beta} - z_i' \hat{\gamma}$
Studentized	$r_i^m / \sqrt{\hat{V}(r_i^m)}$	$r_i^c / \sqrt{\hat{V}(r_i^c)}$
Pearson	$r_i^m / \sqrt{\hat{V}(Y_i)}$	$r_i^c / \sqrt{\hat{V}(Y_i \gamma)}$
Scaled	$\hat{C}^{-1} r^m$	

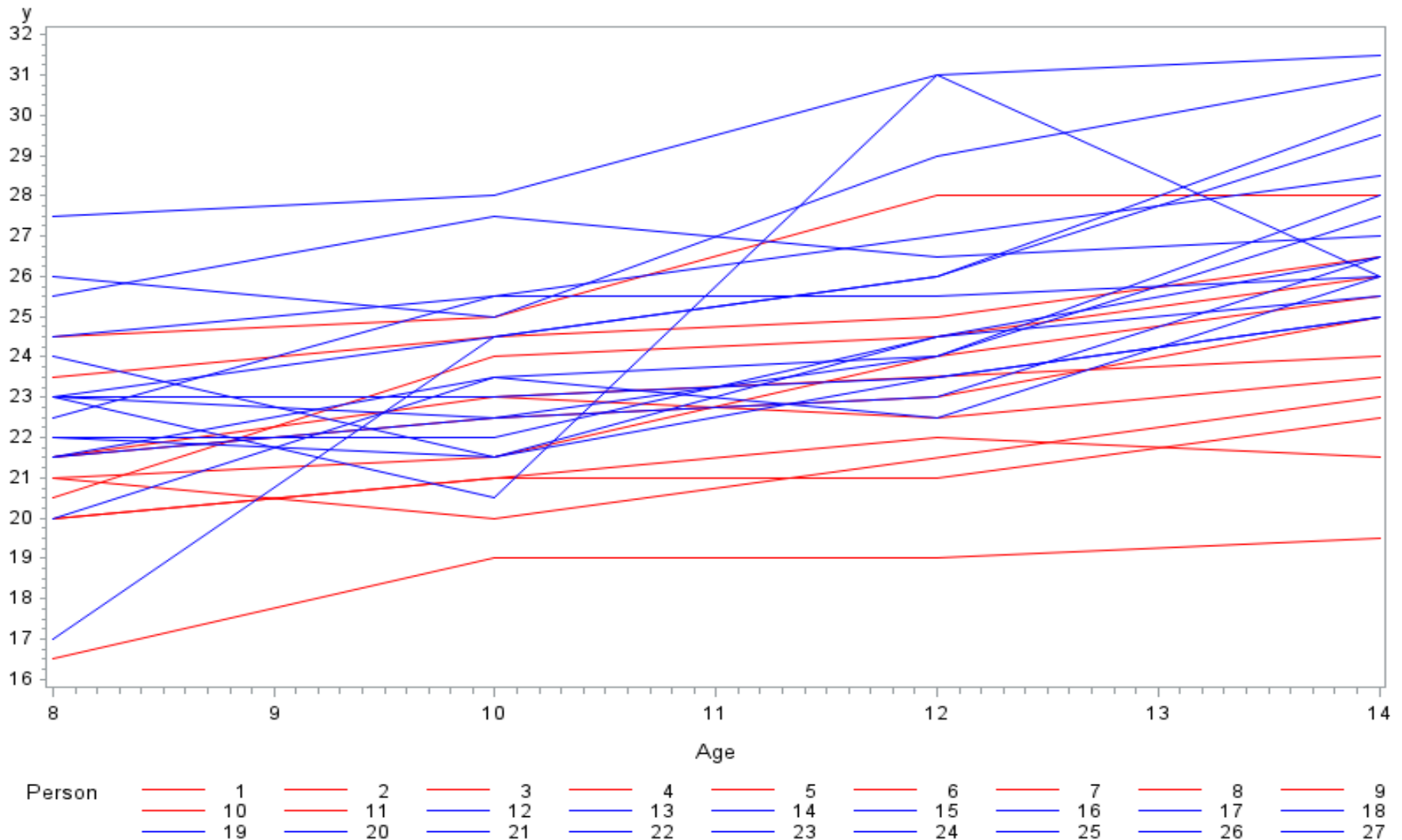
Studentized Residuals

- Scale each raw residual by its estimated SD
 - Internally studentized : obs contributes to standard error computation
 - Externally studentized : estimate of standard error independent of the obs
- Internally available in `outp=` and `outpm=` with residual option
- Externally available with influence option

Scaled residuals

- For correlated data, can consider scaling marginal residuals by Cholesky decomposition of $\text{Var}(Y)$
- Should have mean zero and approximately uncorrelated
- Can be used to diagnose whether the covariance structure is reasonable
- Available with OUTPM= and VCIRY options

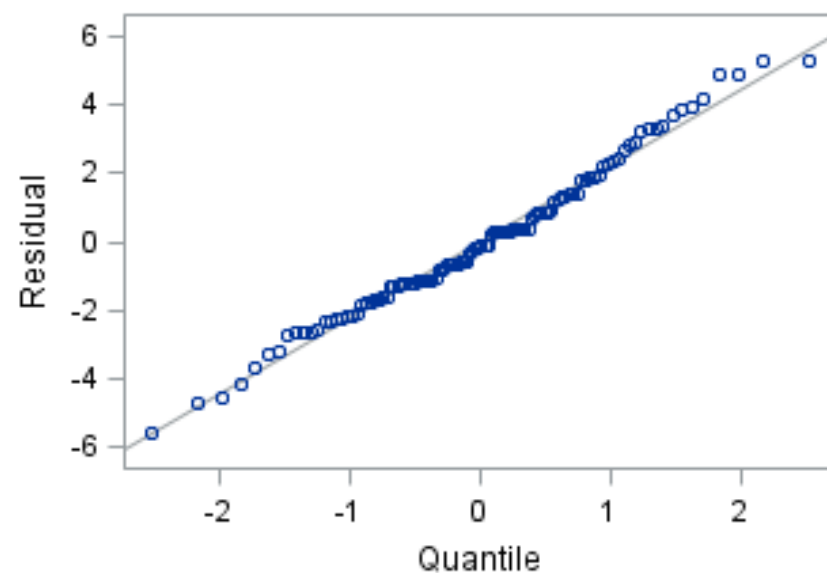
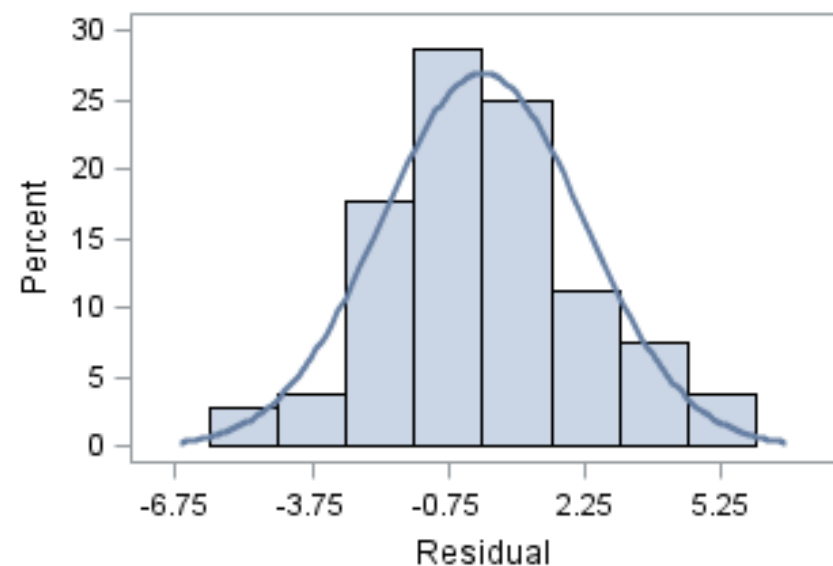
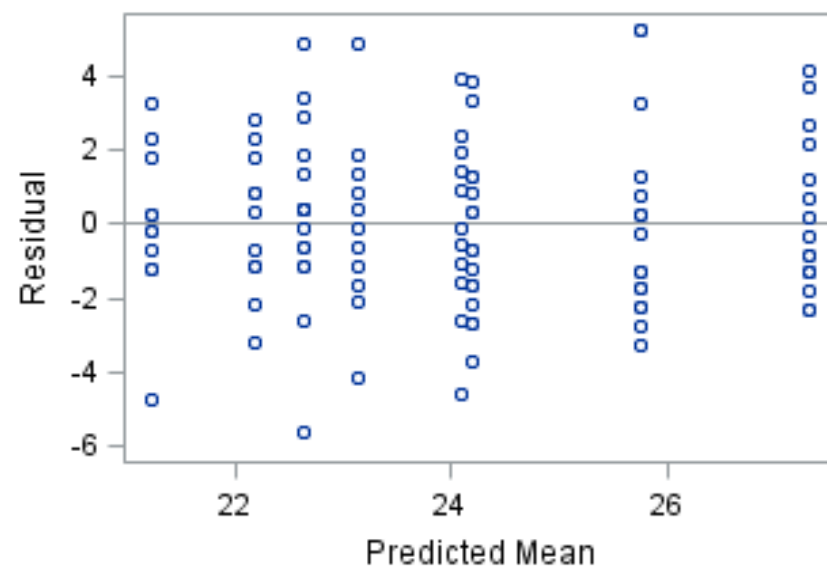
Dentistry Growth Data



SAS Code

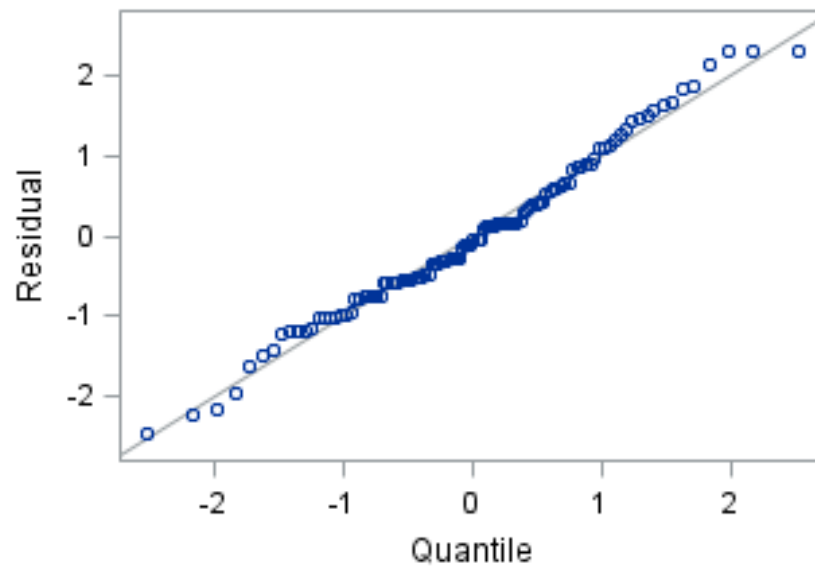
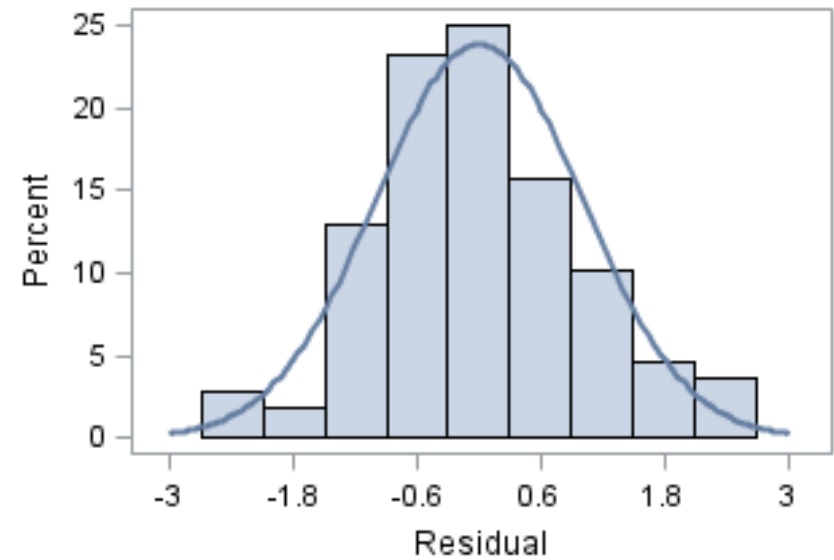
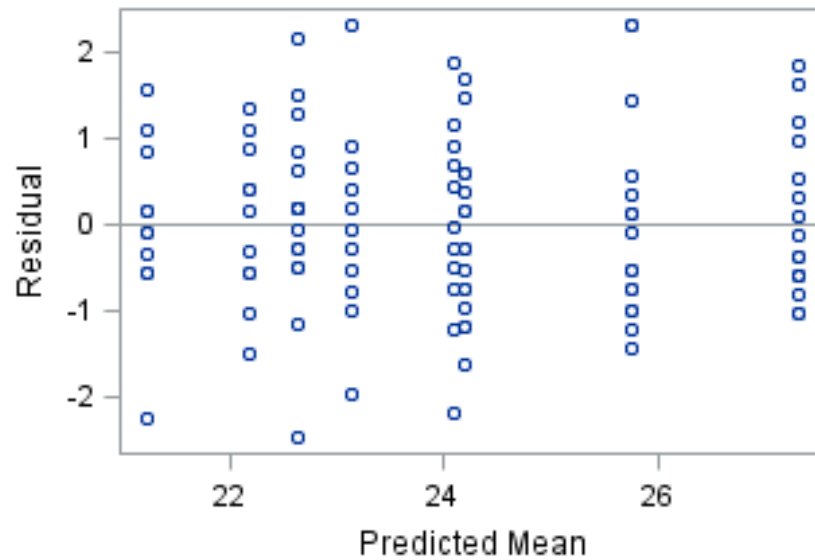
```
proc mixed data=example2 plots=residualpanel;  
  class person sex;  
  model y = sex age1(sex) / noint s ddfm=kr  
                                outpm=resm vciry residual;  
  repeated / type=cs sub=person group=sex r=1,12;  
  estimate 'diff in ints' sex 1 -1;  
  estimate 'diff in slopes' age1(sex) 1 -1;  
run;
```

Residuals for y



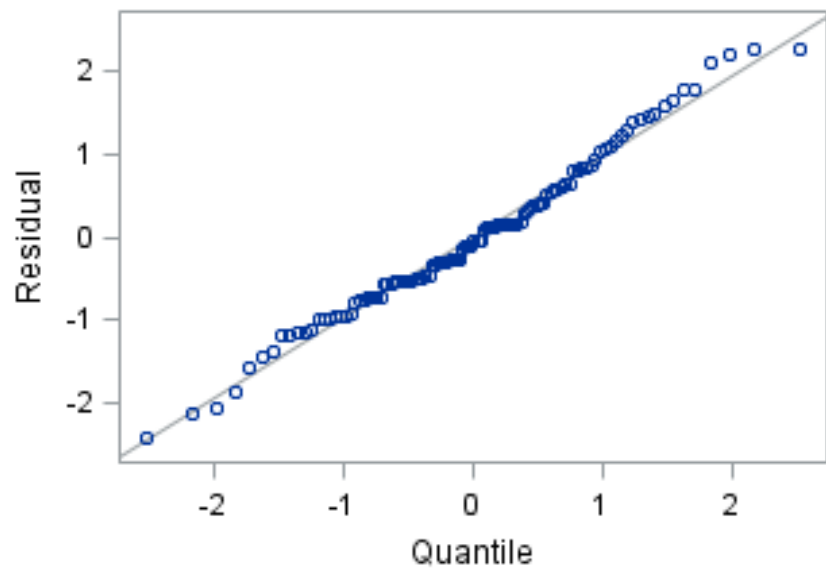
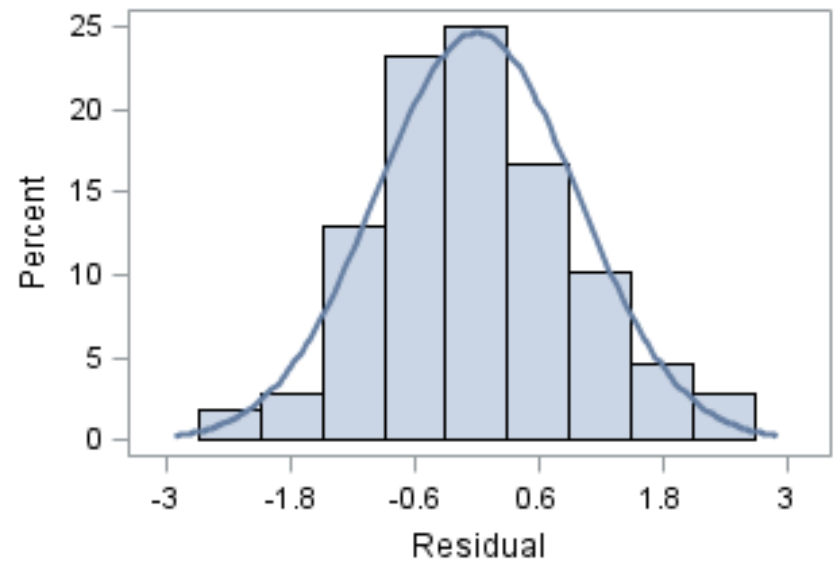
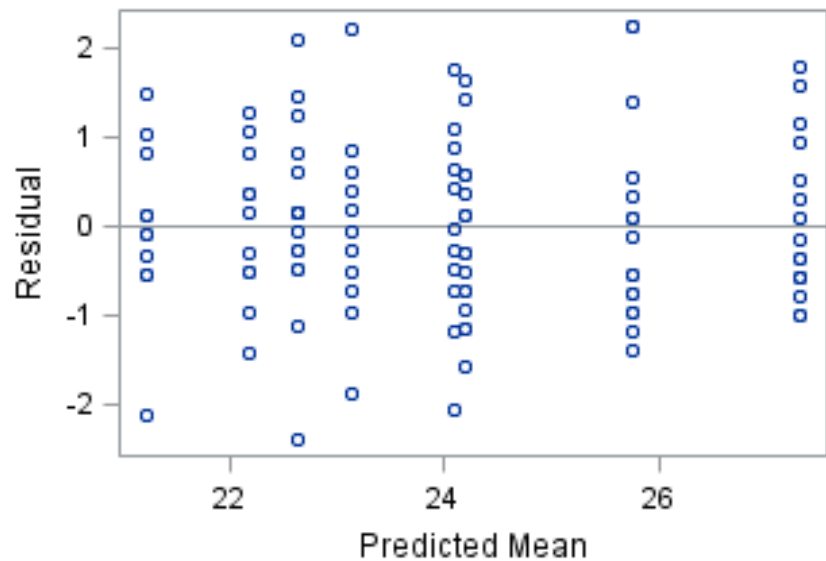
Residual Statistics	
Observations	108
Minimum	-5.616
Mean	-3E-15
Maximum	5.2469
Std Dev	2.2251
Fit Statistics	
Objective	414.67
AIC	422.67
AICC	423.07
BIC	427.85

Studentized Residuals for y



Residual Statistics	
Observations	108
Minimum	-2.47
Mean	0.0002
Maximum	2.3018
Std Dev	1.0046
Fit Statistics	
Objective	414.67
AIC	422.67
AICC	423.07
BIC	427.85

Pearson Residuals for y



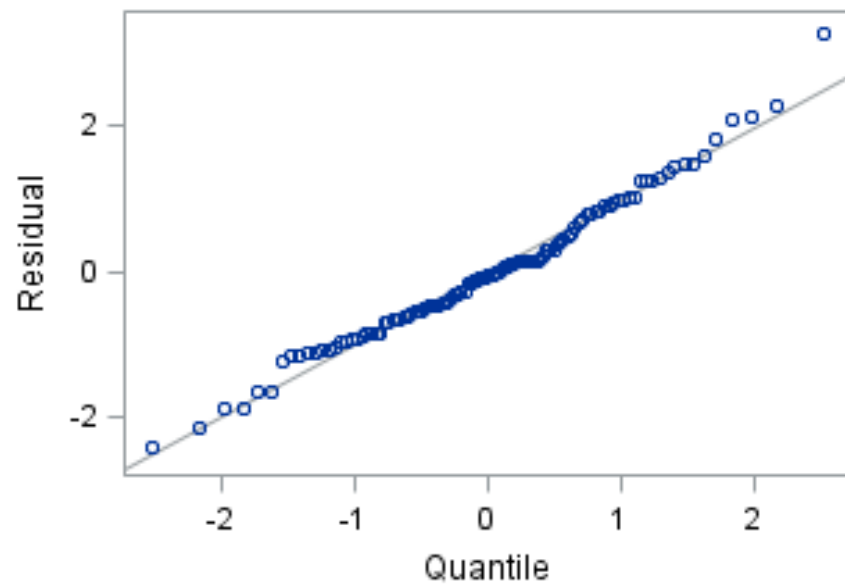
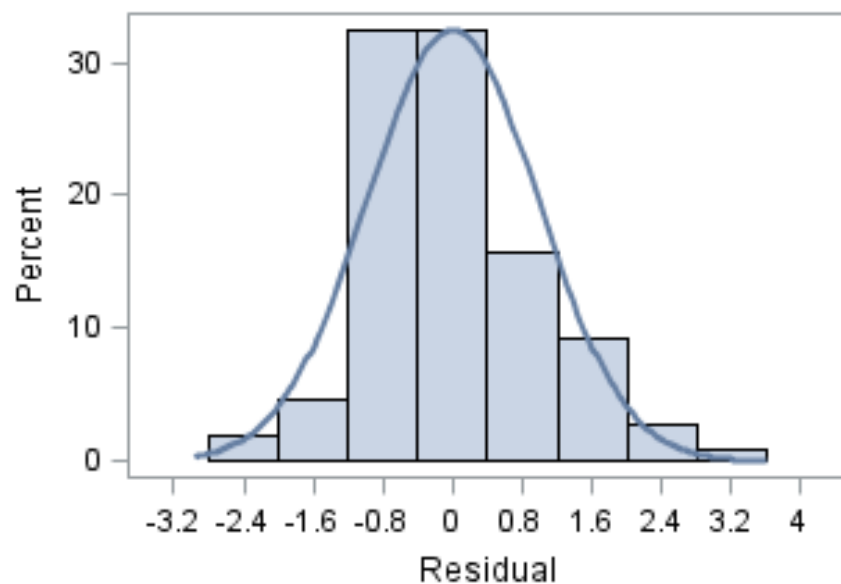
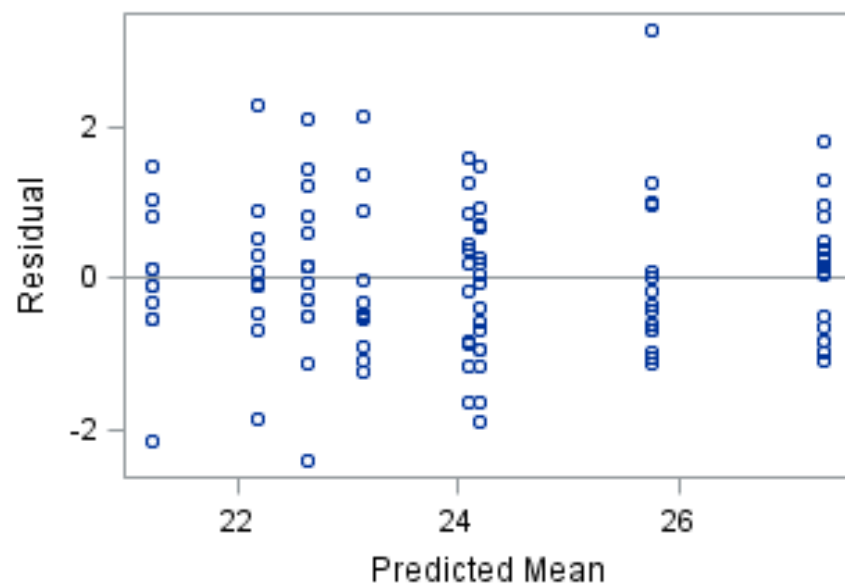
Residual Statistics

Observations	108
Minimum	-2.404
Mean	-1E-15
Maximum	2.246
Std Dev	0.9729

Fit Statistics

Objective	414.67
AIC	422.67
AICC	423.07
BIC	427.85

Scaled Residuals for y



Residual Statistics

Observations	108
Minimum	-2.404
Mean	-0.003
Maximum	3.2718
Std Dev	0.9859

Fit Statistics

Objective	414.67
AIC	422.67
AICC	423.07
BIC	427.85

SAS Code

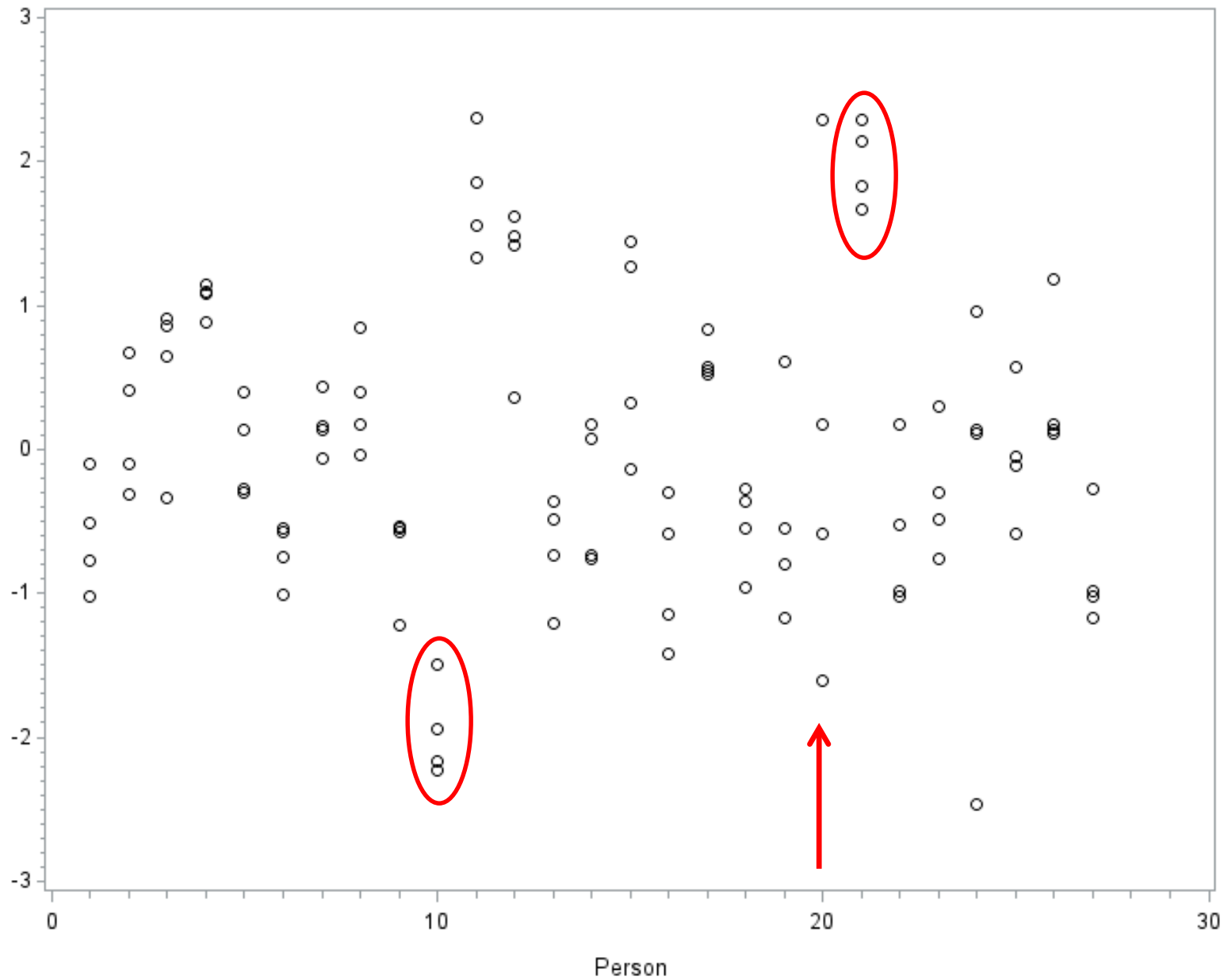
```
proc mixed data=example2 plots=residualpanel;  
  class person sex;  
  model y = sex age1(sex) / noint s ddfm=kr  
                                outpm=resm vciry residual;  
  repeated / type=cs sub=person group=sex r=1,12;  
  estimate 'diff in ints' sex 1 -1;  
  estimate 'diff in slopes' age1(sex) 1 -1;  
run;
```

SAS Code

```
symbol1 i=none v=circle c=black;  
proc gplot data=resm;  
  plot(resid studentresid scaledresid)*person;  
run;
```

Must use data set to look at alternative residual
plots

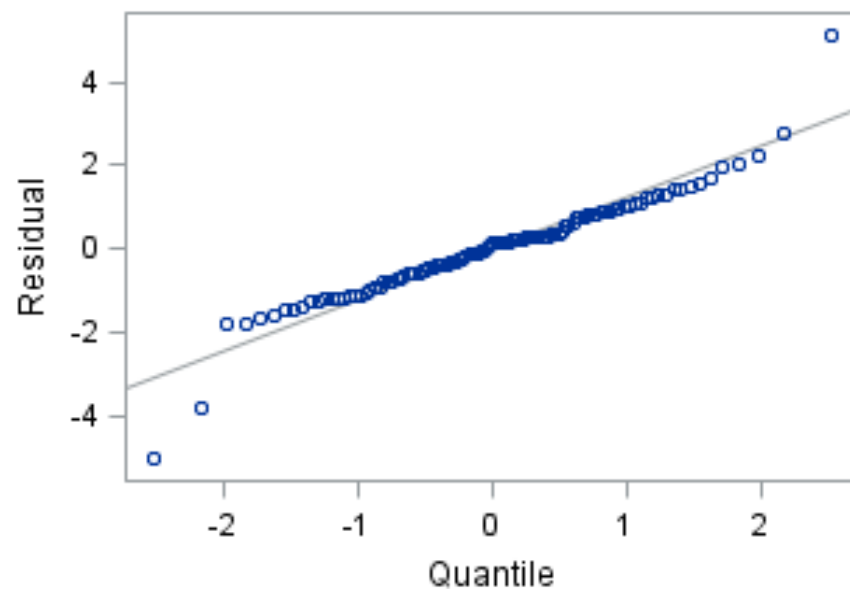
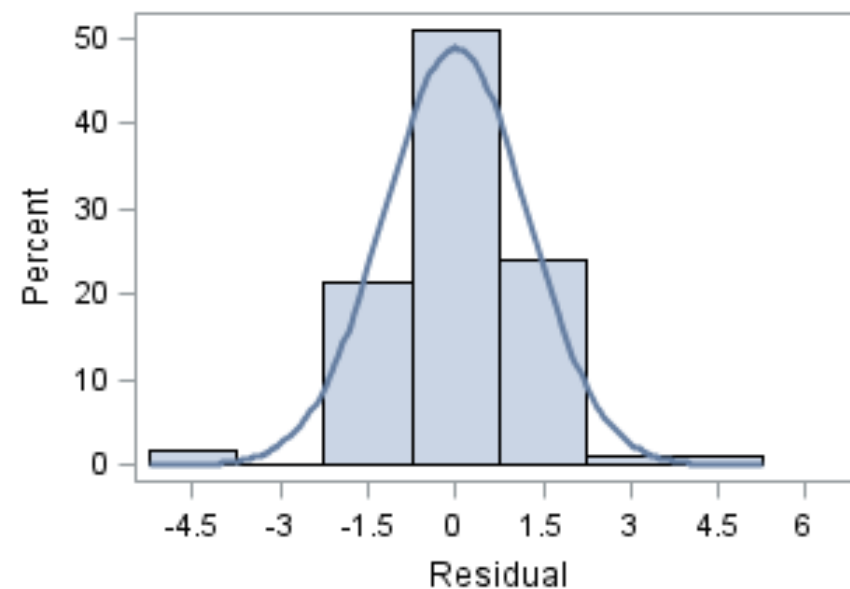
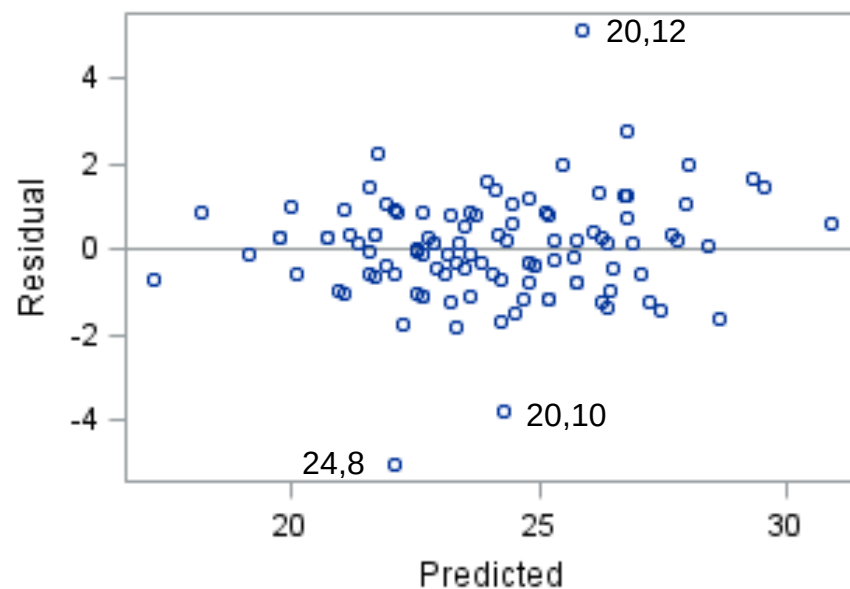
Studentized Residual



SAS Code

```
proc mixed data=example2
    plots=residualpanel(conditional marginal);
class person sex;
model y = sex age1(sex) / noint s ddfm=kr
    outp=resm1 vciry residual;
random intercept / subject=person group=sex;
repeated / sub=person group=sex r=1,12;
estimate 'diff in ints' sex 1 -1;
estimate 'diff in slopes' age1(sex) 1 -1;
run;
```

Conditional Residuals for y



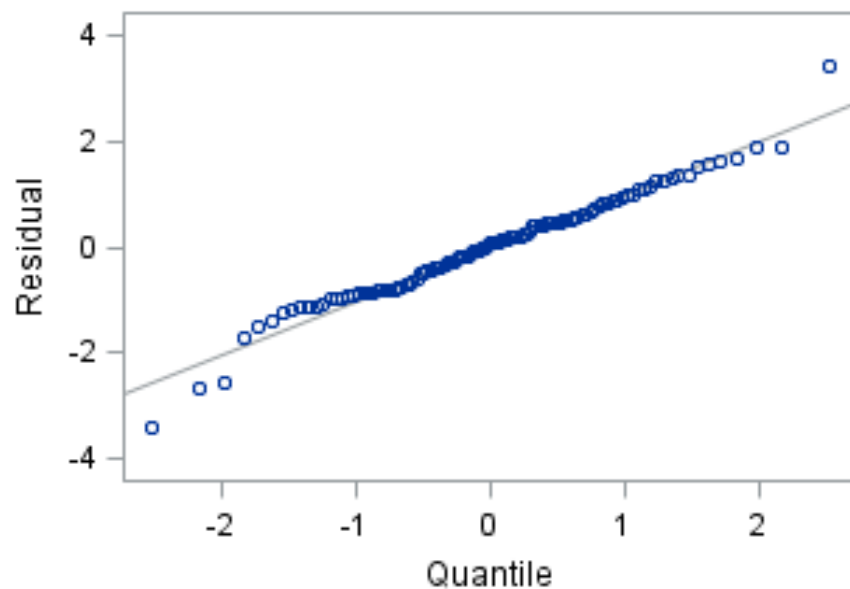
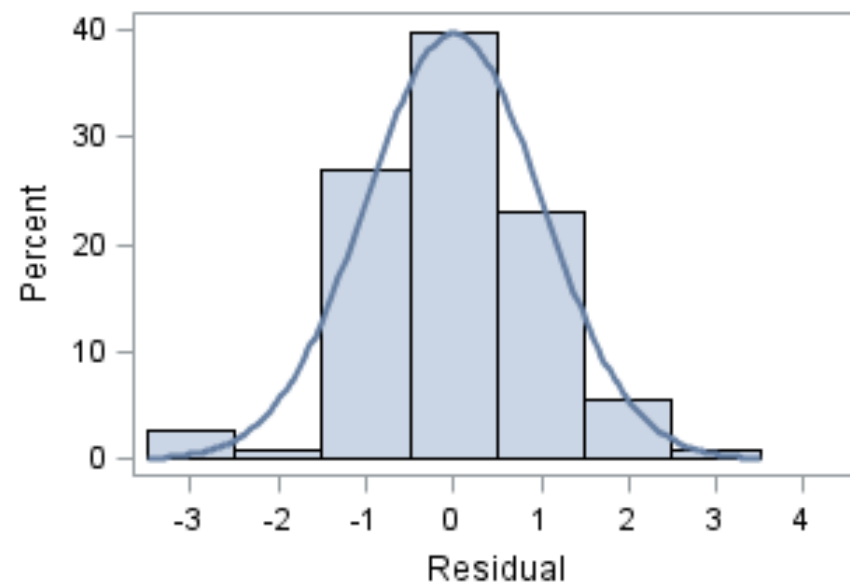
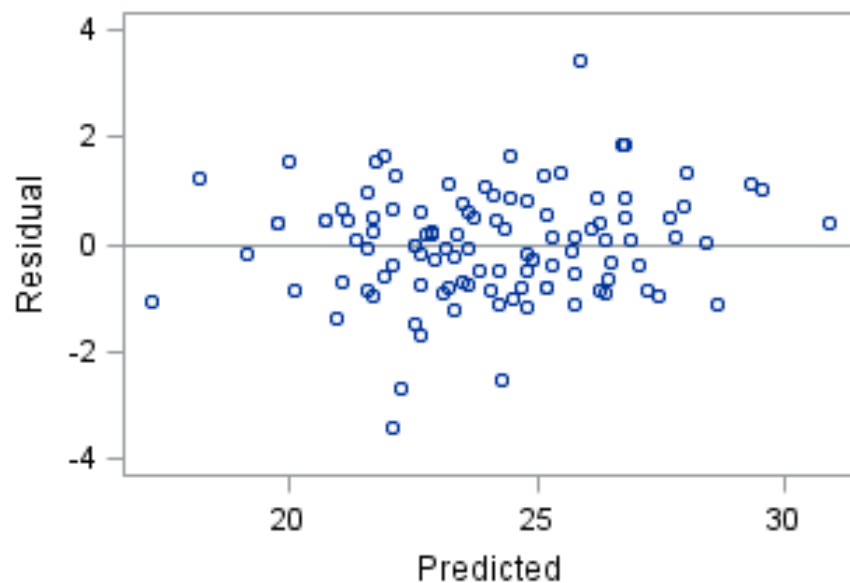
Residual Statistics

Observations	108
Minimum	-5.048
Mean	65E-16
Maximum	5.1235
Std Dev	1.2264

Fit Statistics

Objective	414.67
AIC	422.67
AICC	423.07
BIC	427.85

Conditional Studentized Residuals for y



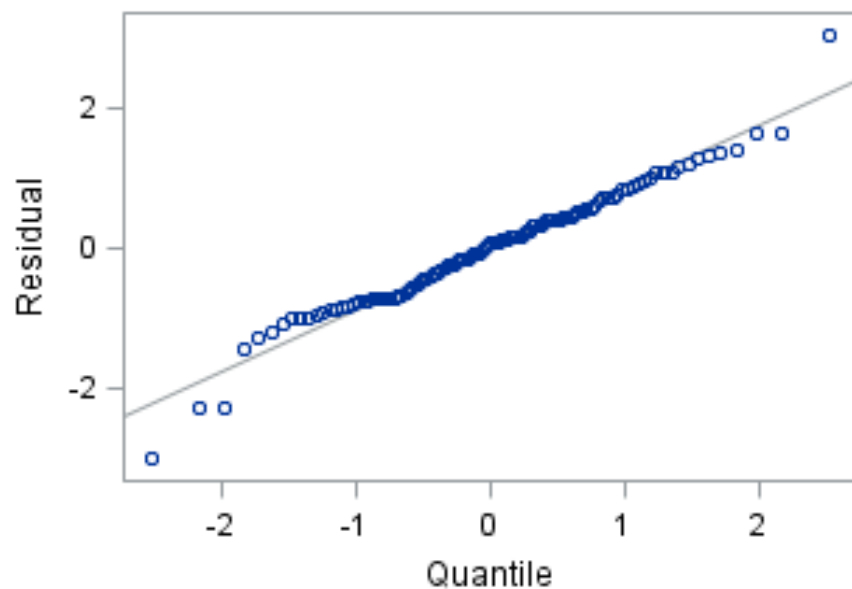
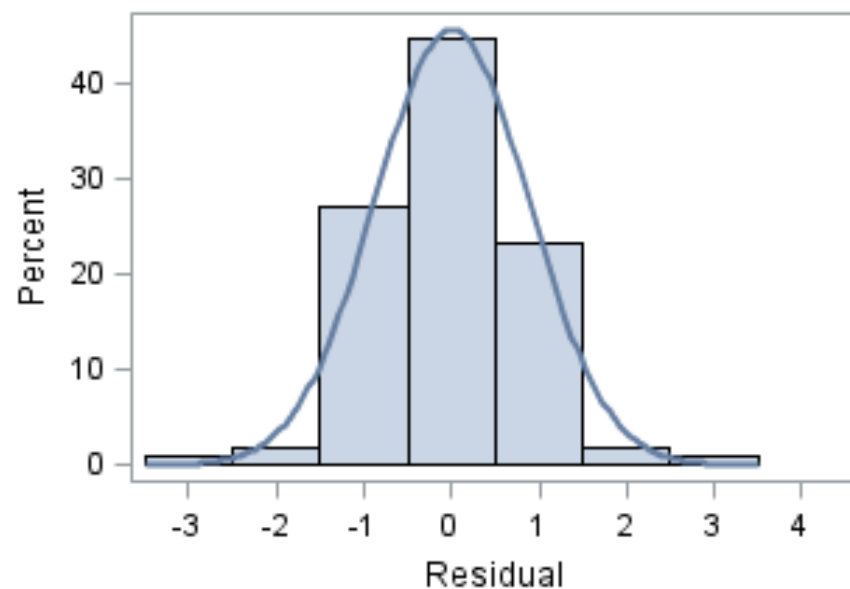
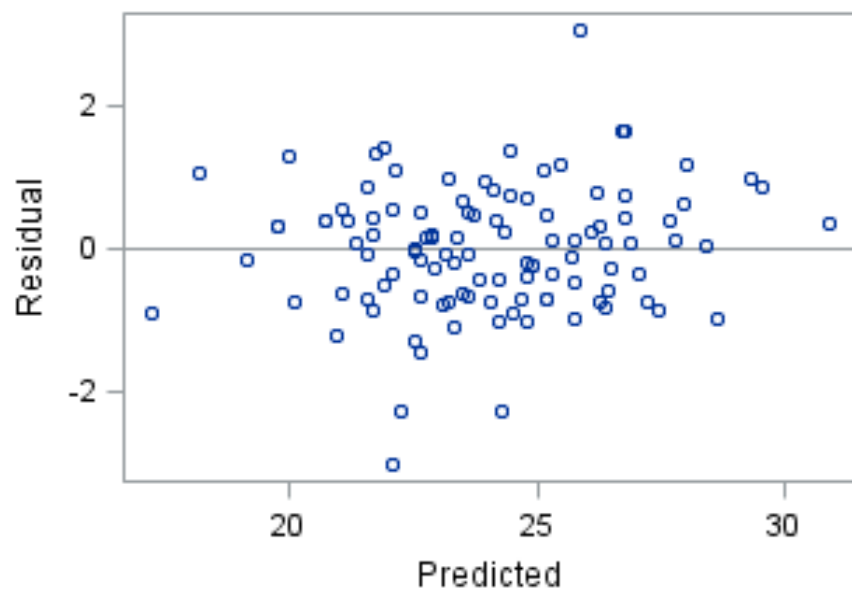
Residual Statistics

Observations	108
Minimum	-3.425
Mean	0.0006
Maximum	3.4214
Std Dev	1.0055

Fit Statistics

Objective	414.67
AIC	422.67
AICC	423.07
BIC	427.85

Conditional Pearson Residuals for y



Residual Statistics

Observations	108
Minimum	-3.008
Mean	64E-16
Maximum	3.0529
Std Dev	0.8781

Fit Statistics

Objective	414.67
AIC	422.67
AICC	423.07
BIC	427.85

Influence Diagnostics

- Basic procedure
 - Fit model and obtain all estimates
 - Remove one or more data points and recompute estimates
 - May not be possible (G not positive definite)
 - Contrast quantities of interest to see which observations are influential

Global Influence Summary

- Likelihood distance (or restricted likelihood)

$$LD_{(-U)} = 2(L(\theta) - L(\theta_{(-U)}))$$

- Measures overall change in fit
 - Likelihood based on all the data, not subset
- If global measure suggests influence, must determine the nature of influence

Types of Influence

- Affects the fixed parameter estimates
 - SAS provides modified versions of Cook's D and multivariate DFFITS

- $$D(\beta) = \frac{(\hat{\beta} - \hat{\beta}_{(-U)})' \hat{v}(\hat{\beta})^{-1} (\hat{\beta} - \hat{\beta}_{(-U)})}{\text{rank}(X)}$$

- $$\text{MDFFITS} = \frac{(\hat{\beta} - \hat{\beta}_{(-U)})' \hat{v}(\hat{\beta}_{(-U)})^{-1} (\hat{\beta} - \hat{\beta}_{(-U)})}{\text{rank}(X)}$$

- Concerned about large values
 - Similar statistics for covariance parameters but do not divide by a matrix rank

Types of Influence

- Affects parameter estimate precision
 - SAS provides summaries of the trace and determinants of the variance matrices
- $\text{COVTRACE}(\beta) = |\text{trace}(\hat{v}(\hat{\beta})^{-1} - \hat{v}(\hat{\beta}_{(-U)})) - \text{rank}(X)|$
- $\text{COVRATIO}(\beta) = \frac{\det_{ns}(\hat{v}(\hat{\beta}_{(-U)}))}{\det_{ns}(\hat{v}(\hat{\beta}))}$
 - Benchmarks are 0 and 1 respectively
 - Similar statistics for covariance parameters except subtract off $\text{rank}(\text{var}(\hat{\gamma}))$

Types of Influence

- Affects the predicted values
 - SAS provides the PRESS residual and DFFITS

$$\text{PRESS}_{(-U)} = \sum (y_i - x_i' \hat{\beta}_{(-U)})^2$$

$$\text{DFFITS}_i = (x_i' \hat{\beta} - x_i' \hat{\beta}_{(-U)}) / \text{ese}(x_i' \hat{\beta})$$

- Leverage – two “Hat” matrices

$$H_1 = X(X' \hat{\Sigma}^{-1} X)^{-1} X' \hat{\Sigma}^{-1}$$

$$H_2 = C^{-1} X(X' \hat{\Sigma}^{-1} X)^{-1} X' C'^{-1} \text{ where } CC' = \hat{\Sigma}$$

Assessment of Influence

- Outlier Detection
 - Studentized residuals within ± 2
- Problem: Iterative analysis approach can be very time consuming
 - Can use full parameter values as starting pts

Influence Options

- INFLUENCE

- (<EFFECT=effect> - effect by which obs are grouped
- ESTIMATES|EST> - updated estimates written to ODS
- <ITER=number> - max number of iterations in MIXED
- <KEEP=number> - how many groups to output
- <SELECT=value-list> - restrict analysis to subset of obs
- <SIZE=number>) - size of groups to remove

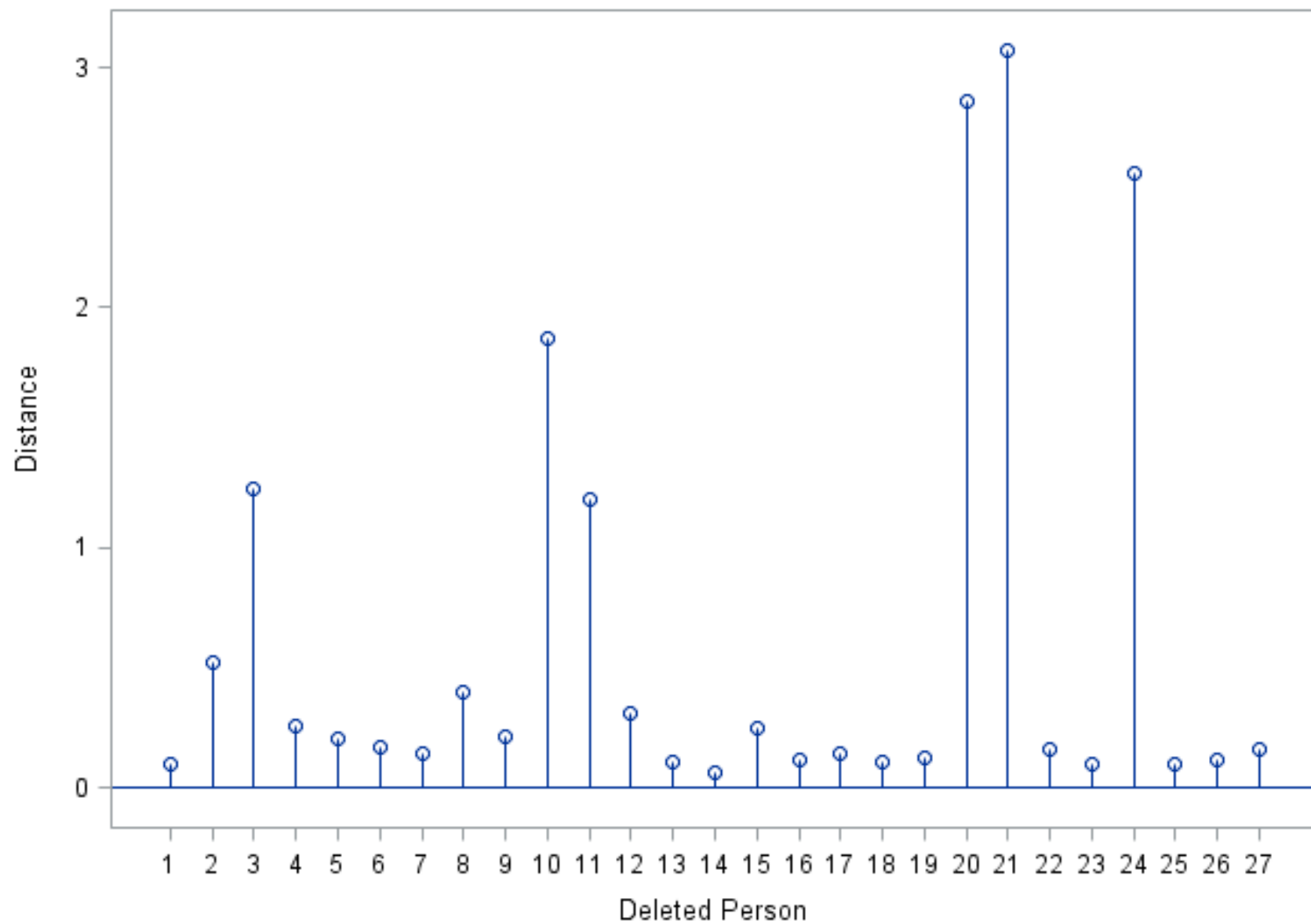
SAS Code

```
proc mixed data=example2;  
  class person sex;  
  model y = sex age1(sex) / noint s ddfm=kr  
    influence(effect=person iter=6);  
  random intercept / subject=person group=sex;  
  repeated / sub=person group=sex r=1,12;  
  estimate 'diff in ints' sex 1 -1;  
  estimate 'diff in slopes' age1(sex) 1 -1;  
run;
```

Influence Diagnostics for Levels of Person

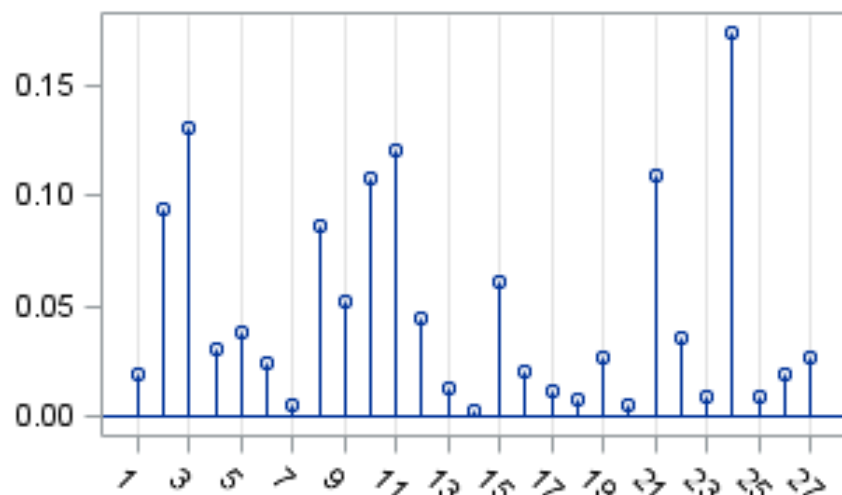
Person	Number of Observations in Level	Iterations	PRESS Statistic	Cook's D	MDFFITS	COVRATIO	COVTRACE	Cook's D CovParms	MDFFITS CovParms	COVRATIO CovParms	COVTRACE CovParms	Restricted Likelihood Distance
1	4	2	9.9795	0.01993	0.01770	1.2690	0.2549	0.02631	0.02146	1.3483	0.3306	0.1025
2	4	3	3.7858	0.09360	0.09081	1.2542	0.2478	0.12405	0.11091	1.3172	0.3295	0.5223
3	4	4	11.0018	0.13159	0.14218	1.0827	0.0987	0.49072	0.62879	0.9816	0.0510	1.2419
4	4	3	24.1855	0.03081	0.02875	1.2891	0.2745	0.14320	0.10916	1.3916	0.3741	0.2540
5	4	3	1.6900	0.03782	0.03394	1.3619	0.3365	0.06443	0.04739	1.5533	0.5041	0.2056
6	4	2	11.8620	0.02427	0.02090	1.3537	0.3271	0.08605	0.06862	1.5342	0.4776	0.1710
7	4	3	1.1708	0.00526	0.00444	1.4365	0.3974	0.14037	0.10753	1.7280	0.6308	0.1452
8	4	2	4.8795	0.08712	0.08021	1.3045	0.2876	0.05048	0.03866	1.4238	0.4013	0.3982
9	4	2	12.9100	0.05221	0.04643	1.2835	0.2662	0.01382	0.01151	1.3795	0.3506	0.2174
10	4	4	84.7368	0.10757	0.15321	0.7857	0.1777	0.66723	1.47375	0.5169	0.4015	1.8738
11	4	3	69.7395	0.12076	0.14018	0.8513	0.1361	0.38820	0.64646	0.6068	0.3805	1.1979
12	4	3	40.5287	0.04458	0.04622	1.0446	0.0488	0.09766	0.10953	0.9646	0.0179	0.3074
13	4	2	13.7370	0.01351	0.01250	1.2115	0.2018	0.06149	0.05224	1.2976	0.2796	0.1110
14	4	2	6.3780	0.00289	0.00256	1.2486	0.2350	0.05656	0.04826	1.3781	0.3485	0.0642
15	4	2	22.5094	0.06146	0.05787	1.1382	0.1339	0.00228	0.00215	1.1454	0.1409	0.2492
16	4	2	21.7880	0.02114	0.02014	1.1575	0.1524	0.03051	0.02687	1.1844	0.1793	0.1152
17	4	2	9.3472	0.01202	0.01093	1.2442	0.2312	0.10576	0.08756	1.3683	0.3409	0.1441
18	4	2	8.2156	0.00793	0.00716	1.2471	0.2336	0.08528	0.07185	1.3748	0.3453	0.1100
19	4	2	15.1330	0.02752	0.02571	1.1784	0.1718	0.01615	0.01398	1.2276	0.2191	0.1250
20	4	3	43.1130	0.00472	0.00607	0.8792	0.0881	1.85934	3.30617	0.6833	0.2153	2.8585
21	4	3	94.7213	0.10928	0.16959	0.7267	0.2276	1.25641	3.08161	0.4668	0.4142	3.0727
22	4	2	13.5948	0.03624	0.03305	1.2041	0.1947	0.02428	0.02149	1.2819	0.2644	0.1644
23	4	2	5.6591	0.00862	0.00771	1.2562	0.2417	0.06808	0.05788	1.3949	0.3625	0.0968
24	4	3	40.4687	0.17427	0.20960	0.9358	0.0418	1.20393	1.85155	0.7742	0.1534	2.5636
25	4	2	3.7459	0.00854	0.00770	1.2662	0.2508	0.07370	0.06189	1.4170	0.3820	0.1015
26	4	2	8.2530	0.01901	0.01718	1.2373	0.2248	0.04651	0.04007	1.3533	0.3271	0.1173
27	4	2	20.3400	0.02754	0.02585	1.1712	0.1655	0.05403	0.04638	1.2125	0.2065	0.1613

Restricted Likelihood Distance

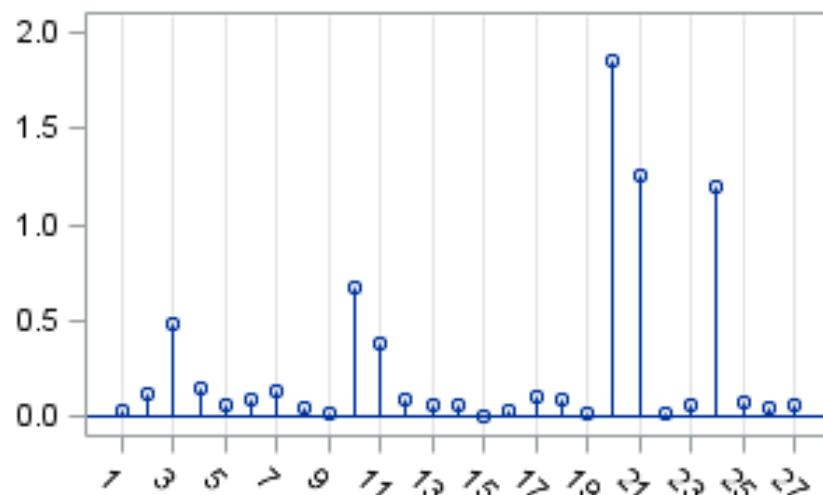


Influence Statistics for y

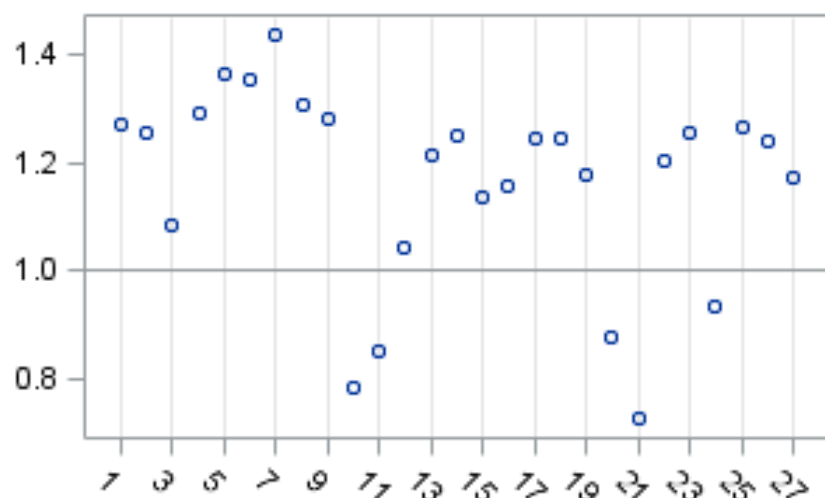
Cook's D Fixed Effects



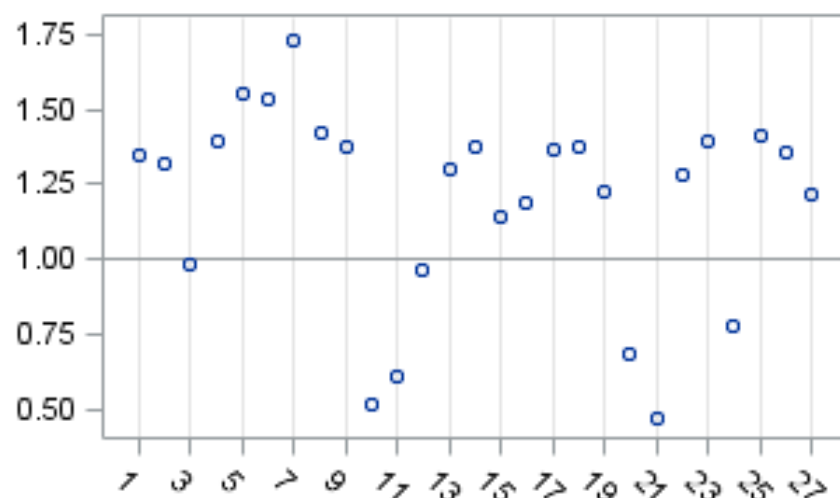
Cook's D Covariance Parameters



CovRatio Fixed Effects



CovRatio Covariance Parameters



Deleted Person

Reference

- Schabenberger, O. (2004) **Mixed Model Influence Diagnostics**, SUGI 29 – Statistics and Data Analysis, **Paper 189-29.**